

Question 1: [10 pts] What is the purpose of the window mechanism in TCP?

It helps manage the flow of packets. By having a window mechanism, receivers are able to communicate the space they have and senders don't overwhelm the receivers, so there is less data loss and more efficiency.

Question 2: [25 pts] Let's observe how the TCP protocol operates. Begin by opening Wireshark and listening on the 'any' interface. Open a terminal window. Using wget, download the file <http://ipv4.download.thinkbroadband.com/10MB.zip>

2.a [15 pts] Find the packets corresponding with the SYN, SYN-ACK, and ACK that initiated the TCP connection for this file transfer. Take a screenshot of these packets and show that. What was the initial window size that your computer advertised to the server? What was the initial window size that the server advertised to you? What are sequence numbers and acknowledgement numbers?

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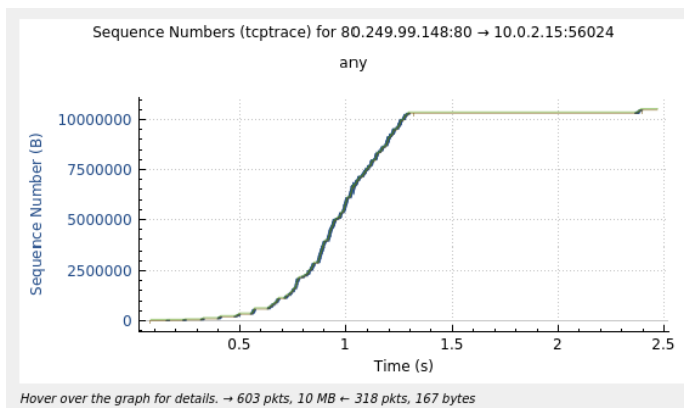
80.249.99.148      TCP      76 56024 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=
10.0.2.15         TCP      62 80 → 56024 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
80.249.99.148      TCP      56 56024 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
80.249.99.148      HTTP     223 GET /10MB.zip HTTP/1.1
10.0.2.15         TCP      62 80 → 56024 [ACK] Seq=1 Ack=168 Win=65535 Len=0
10.0.2.15         TCP      2936 80 → 56024 [ACK] Seq=1 Ack=168 Win=65535 Len=2880 [TCP segm
80.249.99.148      TCP      56 56024 → 80 [ACK] Seq=168 Ack=2881 Win=63360 Len=0
10.0.2.15         TCP      4376 80 → 56024 [ACK] Seq=2881 Ack=168 Win=65535 Len=4320 [TCP
80.249.99.148      TCP      56 56024 → 80 [ACK] Seq=168 Ack=7201 Win=60480 Len=0

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(this was the most I could fit on the screen since it won't let me expand the window)

The initial window size that my computer advertised was 64240 bytes and the initial size that the server advertised to my computer was 65535 bytes. The sequence numbers are 0, 1, 1, 168, 2881, and 168 and the acknowledgement numbers are 1, 1, 168, 168, 2881, 168, and 7201.

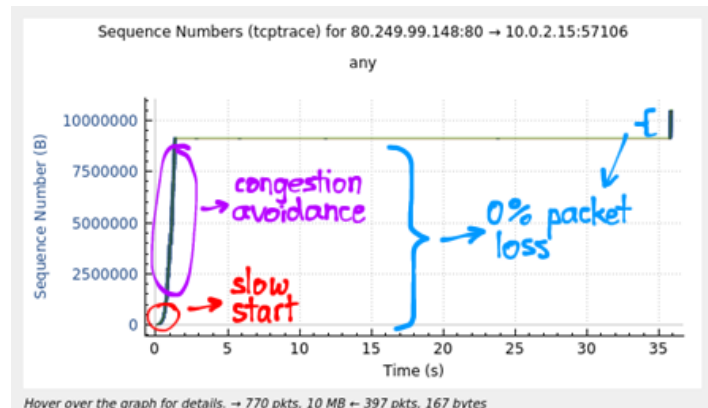
2.b [5 pts] Find a packet from the download with a source of the server and the destination of your computer. Create a tcptrace graph of this packet selected. Take a screenshot of the graph and explain what it is showing. Look into the Wireshark documentation if you need assistance making this graph.



This is a graph with the source being the server and the destination being my computer. It is showing the sequence number changing over the period of a few seconds, which means a little over 10,000,000 bytes were transferred from the server to my computer in a little less than 2.5 seconds. There is a period of no data being transferred at about 1.3 seconds, probably due to my spotty WiFi.

2.c [5 pts] This time simulate packet loss while downloading the zip file. Find a packet from the download with the source of the server and the destination of your computer. Create a tcptrace graph of this packet selected. Take a screenshot of the graph and explain what it is showing. Using an image editing program, circle the areas where the 0% loss is shown, as well as where TCP is in slow-start and congestion-avoidance.

This is, similar to the last image, showing the server sending data to my computer. In this one, there is a long packet loss period (about 30s because that was the fastest I could run the commands) before it fully sends everything. The slow start is the exponential part, the linear growth is congestion avoidance, and anywhere not a flat line is 0% packet loss.



Question 3: [25 pts] Suppose you are in charge of the address space 224.63.17.0/24.

3.a [3 pts] Write the number of bits for the network and host portions.

24 network bits and $32 - 24 = 8$ host bits

3.b [4 pts] How many hosts can be assigned addresses in this original network?

$2^8 - 2 = 256 - 2 = 254$ host bits

3.c [4 pts] Suppose this network is too big and you decide you would like to divide your network into 32 subnets. How many bits are needed to make 32 subnets?

$2^x = 32$ so $x = 5$ bits

3.d [3 pts] How many hosts can be in each subnet?

$8 - 5 = 3$ so $2^3 - 2 = 8 - 2 = 6$ hosts

3.e [6 pts] Write the first 3 subnet addresses in dotted decimal form.

224.63.17.0, 224.63.17.8, 224.63.17.16

3.f [5 pts] Write the IP address for all of the hosts that can be connected to the 1st subnet in dotted decimal notation.

224.63.17.1, 224.63.17.2, 224.63.17.3, 224.63.17.4, 224.63.17.5, 224.63.17.6

Questions 4: [10 pts] Please list the advantages and disadvantages of introducing Network Address Translation (NAT) technique.

pros: allows multiple devices to share a public IP address so they don't all need to immediately switch to IPv6, provides a layer of security, preserves intern IP addresses if an organization switches ISPs, allows for devices to only need one instead of multiple IP addresses

cons: alters IP headers so might not be compatible with a receiver, makes error checking harder, requires extra processing and resources

Question 5 [20 pts]: Given the following destination IP addresses, determine which next hop each packet should be forwarded to:

5.a [4 pts] 10.1.2.130 3

5.b [4 pts] 10.1.2.70 2

5.c [4 pts] 10.1.3.5 1

5.d [4 pts] 192.168.10.10 4

5.e [4 pts] 10.1.2.130 3

Question 6 [10 pts]: Given mixed IPv4 and IPv6 routers, how does network operate with it to support data transmission?

The IPv6 packet is encapsulated inside an IPv4 packet (by modifying IP headers) at the first IPv6/v4 server, which then travels through the IPv4 servers, then gets unpacked in the second IPv6/v4 server (by modifying IP headers) and is transferred to the last IPv6 server.